

ALLEGRI (Artificial Intelligence-assisted Reading of Gram Stains and Blood Smears)

Contribution of Artificial Intelligence to the Interpretative Reading of Gram Stains and Blood Smears/Thick Blood Films.

Brief description:

Bacterial and parasitic infections, particularly malaria, are leading causes of worldwide mortality where the patient's prognosis depends directly on the speed of treatment. While traditional diagnosis relies on direct microscopic examination, expert interpretation is often unavailable 24/7, especially during night shifts. Project ALLEGRI introduces an AI-assisted system coupling a high-fidelity digital camera integrated into a microscope with a web-based application. By automating the identification of pathogens in Gram stains and blood smears, the device aims to secure and standardize medical interpretation. This solution offers a decisive time-saving advantage for real-time diagnostic support, ensuring faster and more reliable patient care in clinical environments.

Team Number : n°498

Project Partner : Delafontaine Saint-Denis Hospital

Team Member	Role
CATTEAU Camille	Image retrieval
MERCIER Thomas	Image retrieval
BOUCHENTOUF Othman	IA identification model
SHAN Coralia	Application development
BOLLU Shubham Nagnath	Application development
O'KELLY Maxence	Team referent
MELLOULI Nedra	School referent
Hassold Nolan	Hospital referent

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1 Objectives Set versus Results Achieved

1.1 Project Initial targets

The initial objectives of the ALLEGRI project were defined during the kickoff meeting with our hospital partner, Nolan Hassold from the Delafontaine Saint-Denis Hospital. No strict quantitative targets (such as a specific accuracy threshold, formal budget, or fixed delivery date) were set at this stage. The objectives were expressed in qualitative terms, focused on producing a functional proof-of-concept by the end of the academic year.

Three main goals were established:

- AI objective : Develop a model capable of automatically identifying pathogens in two types of microscopic samples, Gram stains for bacterial detection, and blood smears for malaria detection. The system should be reliable enough to serve as a real-time decision-support tool for clinical staff, particularly during night shifts.
- Hardware objective: Design and build a low-cost image acquisition device that can be mounted on a microscope to capture high-quality images and transmit them to the AI model without requiring expensive external equipment.
- Software objective : Develop a web or mobile application allowing medical staff to interact with the system, visualize detection results, and validate them easily from any hospital device.

The objectives did not change significantly throughout the project. The overall scope remained consistent with the initial brief, and the partner's expectations stayed aligned with our deliverables.

1.2 Results Achieved

The project successfully delivered on its three core objectives, though with some nuances depending on the component.

Positive outcomes:

The AI model performed strongly in training conditions, reaching a mAP50 of 92.9% for bacteria and 87.9% for malaria detection. The results clearly meet the goal of building a reliable diagnostic support tool.

The web application was also a clear success : it provides an intuitive interface for medical staff to capture images, visualize detection results with bounding boxes and confidence scores in real time, and interact with the system from any device.

Some improvements are still planned, such as adding a permanent session history, but the core functionality works well and was positively received by the partner.

Negative deviations and limitations:

The main limitation of the project is the gap between training performance and real-world reliability. When tested against actual diagnoses from hospital doctors, the model's predictions were not always accurate. This is directly linked to our inability to test and train the model on real clinical slides: hospital regulations prevented us from bringing samples to school throughout the year. However, this point evolved positively, during our last meeting, the hospital obtained formal

authorization to allow us to bring labelled slides to school for testing. This will be a key improvement for the next iteration of the project.

On the hardware side, the Raspberry Pi 5 showed processing limitations when running the YOLOv11 model locally, especially on bacteria-dense slides, leading to slow inference times and occasional crashes. Migrating the AI processing to a dedicated cloud server or even a standard laptop CPU would significantly resolve this bottleneck.

Finally, the camera hardware prototype was developed and assembled, but could not be fully validated under real microscope conditions at the hospital. This validation remains an important next step.

Overall, the partner acknowledged the system as a solid and promising proof-of-concept, with constructive feedback focused on scalability, real-world testing, and performance optimization rather than fundamental rework.

1.3 List of Deliverables

The project resulted in a set of complementary deliverables covering hardware, software, and artificial intelligence components. First, a comprehensive technical report was produced, detailing the overall system architecture, the methodology followed, and the different contributions of each team. This includes the design and implementation of the image analysis pipeline, as well as the integration of the various subsystems developed throughout the project.

A functional hardware prototype was developed by one part of the team, consisting of a device integrating a camera and optical lenses designed to capture high-quality images of blood smears. This acquisition system enables reliable data collection and constitutes a critical component of the overall solution, ensuring that the input images are suitable for further automated analysis.

In parallel, a software prototype was implemented to process and analyze the captured images. This includes the development and deployment of the machine learning model used for microorganism detection. The model is hosted on a Raspberry Pi 5, which serves as the central processing unit of the system, handling both inference and communication between components.

Additionally, a web and mobile application was developed by another team, providing an interface for users to interact with the system. This application allows users to control the device, upload or capture images, and visualize the analysis results in real time. It communicates directly with the Raspberry Pi 5, ensuring seamless integration between the user interface and the embedded system.

All source code and related resources were organized and made available through a Microsoft teams repository.

At the time of submission, these deliverables have been presented to the project partner. The overall system has been positively received, with some comments and suggestions for further optimization, particularly regarding performance, scalability, and robustness. These remarks are being considered for future iterations of the project.

Supporting materials and additional documentation are included in the annexes of this report where relevant (Github link).

2 Methodological reviews

The ALLEGRI project was conducted using a decentralized Agile approach, focusing on autonomy and constant adaptation. Instead of a rigid hierarchy, technical decision-making was entrusted to specialized pairs to maintain high velocity.

- **Team Composition and MBTI insights:** The team was organized into three specialized pairs (AI, Image Transmission, and Application). Othman (DIA major) was logically assigned to lead the AI developments due to his technical background. While we did not start with a formal MBTI strategy, a post-project analysis showed that our profiles naturally balanced the team. Each pair had a mix of Introverts/Extroverts, and the presence of at least one "Judging" (J) profile per group (example : Camille, ISFJ) was key to ensuring deadlines were met and preventing information silos.
- **Operational Management (Microsoft Planner):** We managed our workflow through Microsoft Planner using a clear "Bucket" system (see Annex 9.1). Tasks were categorized into specific poles (Interface Utilisateur, Récupération d'images, Développement de l'IA, Échéances), ensuring clear accountability. Every task was assigned to a specific member with defined priority levels, allowing each pair to work autonomously while keeping a global vision of the project's health.
- **Iterative Process and Decision-Making:** We followed a versioning logic (IA v1, Solution v1), which allowed us to build and test functional prototypes quickly before final integration. This decentralized decision-making meant each pair was free to make technical choices (like selecting specific UI components or optical lenses) to speed up development without waiting for top-down approval.
- **Communication and Client Alignment:** Internal synchronization was maintained through seven formal follow-up meetings on Microsoft Teams and constant technical troubleshooting via WhatsApp. A critical part of our methodology was learning to adapt our speech for our client, Nolan, by focusing on clinical results and practical outcomes rather than technical jargon during our updates.

This Agile organization allowed us to stay resilient despite the high pressure of exams and successfully handle major technical pivots, such as the complete change of hardware architecture during the project.

3 Risk Management

We identified some potential risks of this project and actions we organized in order to manage those risks.

- **First risk of the project : lack of organization.**

➔ **Tools implemented** : Different communication channels ; we used a Whatsapp group for timely communication between us, and also to liaise with the school and hospital representatives.

We created an overall plan in Teams at the beginning, which we updated as the project progressed.

We also used Teams to share files. Finally, we managed deliverables and task allocation in Loop.

These methods have been successful, with consistent participation from all group members.

- **Second main risk : Poor management of a professional project.** This was our first “real-world” project, carried out in collaboration with a hospital and intended to be implemented for actual patients. We were not familiar with how to work with healthcare professionals, especially without any dedicated time allocated to the project, with the client.

➔ **Tools implemented** : Regular communication and regular visits to present our progress and test our prototypes. We therefore learned to adapt the work we had initially planned and adapt our communication.

4 Constraints

During this project, we had to take into account many constraints of different kinds.

Regulatory (medical) constraints : We developed an AI model to detect and identify bacteria and parasites. To train our model, we needed medical images. We found some datasets online, but most of them were inaccessible due to privacy concerns, and the publicly available ones were insufficient for our model. Moreover, the images did not match those analyzed in the hospital.

Hospital and school constraints : Authorizations are required to transfer slides from the hospital to the school. Therefore, we were unable to use real samples to test our prototype.

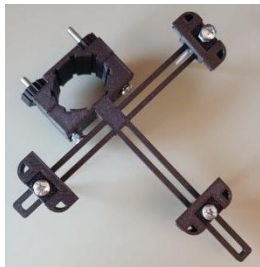
Material constraints : The analyses are performed using a microscope, but the school does not have one. This was another obstacle when trying to test our solution, in addition to the challenges related to image acquisition.

Cost constraints : Some equipment was necessary for our prototype to function, such as a camera and a lens. However, neither the school nor the hospital had a budget allocated for the project, so we had to cover the costs ourselves, and be reimbursed afterward.

5 Technical Review

Technical options :

The technical architecture of the ALLEGRI project was iteratively designed to strictly meet the functional constraints of a clinical laboratory and the client's demands. For the optical acquisition phase, we initially explored a smartphone-based solution, we therefore designed a phone support to be fixed directly on the microscope's ocular. Here's the first prototype of this solution, 3D-printed in PLA :



However, this was quickly discarded as it failed to provide the precise, stable optical focus required for high-resolution microscopic analysis. It was hard to put the phone correctly and really hard to do the correct focus. Consequently, we pivoted to a dedicated digital camera to guarantee high-fidelity image capture. After making a full benchmark, we bought the Bresser MikroCam SF 5.0 MP, in addition to a specific lens "AmScope Lentille de réduction 0,5X" (links in the annex). To meet the functional requirement for a fully autonomous, and space-efficient system, we integrated a Raspberry Pi 5. This choice eliminated the need for a bulky, expensive external PC while providing sufficient local processing power to run the YOLOv11 model securely and in real-time. Finally, to ensure maximum accessibility for medical staff, we developed a web-based application rather than a native software program, allowing biologists to seamlessly review, edit, and validate diagnostic results from any available hospital device.

Outcome of the project :

- **Technical outcomes :** The technical proof-of-concept was highly successful. The end-to-end pipeline operates cohesively : the digital camera captures an image (we still need to validate the precision and validity of the images in real situation, on the binocular microscope of the hospital), the Raspberry Pi 5 autonomously processes the data using the YOLOv11 model without relying on a cloud connection, and the web application instantly displays interactive bounding boxes with confidence scores. By achieving over 92.9% of mAP50 for bacteria and 87.9% mAP50 for malaria, the system successfully fulfills its goal of acting as a reliable, real-time diagnostic safety net for clinical night shifts. However, when comparing results to the Delafontaine hospital doctors diagnostic's, sometimes, results weren't always correct.
- **Organizational outcomes :** Separating our six-person team into three specialized working pairs (AI development, hardware/transmission, and web interface) proved to be a highly efficient organizational strategy. This structure enabled parallel development workflow, expertise in each task and prevented technical bottlenecks. By distributing tasks based on both technical expertise and complementary soft skills, each sub-team maintained high productivity while ensuring their specific modules easily integrated into the whole system architecture.

- **Managerial system outcomes** : The project was driven by a decentralized, agile managerial approach. Rather than employing a top-down hierarchy, with one leader, technical decision-making was entrusted directly to the specialized pairs, which resulted in high team engagement and accountability. We were really able to trust each other on decision-making. Internal cohesion was maintained through regular synchronization meetings and transparent task tracking (via Teams and WhatsApp). Externally, the management strategy successfully adapted to the fast-paced medical sector by maintaining continuous feedback loops with our clinical client (Nolan), via WhatsApp and regular meetings at the hospital, ensuring our engineering deliverables perfectly matched real-world medical expectations.

Dedicated resources :

- **Human resources**: The project was allocated a dedicated team of six engineering students for the academic year. We were 4 MedTech & Health students, 1 Data & Artificial Intelligence student and one Industry & Robotic student. While the total volume of hours was respected, mainly using the Tuesday morning slot to gather and work together, there was a little discrepancy in how these hours were consumed. An unexpected workload spike occurred in the final weeks. The team had to use significant overtime due to the simultaneous arrival of heavy academic deliverables (final reports, soft skills booklets, exams, showroom organization, all in one week).
- **Financial resources**: The dedicated financial budget, allocated specifically for acquiring the Raspberry Pi 5, the digital camera, and the mounting components, was not really defined at the start of the project. We had to justify the need of buying the camera for example. The Raspberry PI was given to us at the beginning of the project. Other potential expenses were avoided by using for example, exclusively open-source software frameworks (YOLO, React, FastAPI), which ensured that absolutely no financial resources were consumed by software licensing or cloud hosting fees.
- **Technical resources (Data & Compute)** : We dedicated raw, high-resolution clinical datasets and GPU compute power. However, the technical time and computing logic consumed during the data preprocessing phase vastly exceeded our initial estimates. We discovered extreme internal discrepancies within the raw data: 70% of the Gram patches were empty backgrounds, and the Malaria dataset had a severe 806:1 healthy-to-parasite ratio. This unexpected reality forced the AI team to consume a lot of technical resources, engineering custom downsampling and dynamic zoom augmentation scripts before model training could even begin.
- **Organizational resources** : We dedicated specific communication infrastructures (one Teams channel, 2 WhatsApp groups) and weekly meetings. The team was divided into three autonomous working pairs, organizational overhead was kept low, and communication resources were consumed efficiently without causing meeting fatigue.

Expected end date vs actual date :

We had a bit of freedom regarding the deadline of the project with our tutor from the hospital. Therefore, we only had to follow the school's deadlines. We managed to have a good group dynamic and work efficiently on the project allowing us to finish in time. A few problems occurred slowing us down but we managed to successfully find solutions to them. We discovered on the way some possible improvements and things to add that we didn't have time to finish, but it exceeded initial expectations.

There were a few times when we found it very difficult to make progress with our project due to communication issues with the 2 tutors and the time it took to get answers to our questions. There wasn't really much we could do about it, other than wait. Because everyone had different schedules, it was sometimes difficult to meet up, which slowed us down a bit, especially when it came to taking turns working on certain physical materials. Furthermore, our client's role at the hospital made it very difficult to arrange visits. Sometimes we would make progress on a solution before realizing that better ones existed, especially on the image transmission part. We lost a lot of time on the mobile solution, before realizing it wouldn't work, once tested on a real microscope. We could probably have anticipated some of them, but they mainly resulted from the various tests we carried out during the process, which made any anticipation difficult. One particular example was collecting slides directly from the hospital. We would have needed real samples in order to test and improve our AI under real-world conditions. Unfortunately, due to health and safety concerns, the hospital was never able to obtain formal authorization to give them to us.

6 Engineering standards

During the development of the ALLEGRI project, our team adhered to several fundamental engineering, software, and hardware standards to ensure the system's reliability, interoperability, and compliance with the medical context:

- **Hardware and optical standards :**
 - **USB Video Class (UVC) :** The standard protocol utilized by the Bresser MikroCam SF 5.0 MP to interface seamlessly with the Raspberry Pi 5 without requiring custom drivers.
 - **Additive Manufacturing Standards :** For our initial smartphone mount prototype, we utilized standard 3D printing protocols (STL file formats and PLA material properties).
- **Software and Mobile Development Standards :**
 - **Android Development Guidelines :** The mobile application was developed adhering to standard Android SDK architectures and UI/UX paradigms (such as Material Design principles) to ensure accessibility for medical staff.
 - **REST API Architecture :** Used to build the communication bridge between the Android application and the FastAPI backend on the Raspberry Pi. This ensured standardized HTTP communication between the mobile interface and the local server.
 - **JSON (JavaScript Object Notation) :** The standard data-interchange format used to transmit YOLOv11 bounding box coordinates and confidence scores to the frontend.
- **Networking and Security Standards :**
 - **IEEE 802.11 (Wi-Fi) & TCP/IP :** The foundational wireless networking protocols used to establish the local connection between the Raspberry Pi and the Android application.
 - **GDPR / RGPD (Privacy by Design) :** Because hospital health and safety regulations prevented the transfer of real clinical slides, we designed our initial prototype to run the YOLOv11 model locally on the Raspberry Pi 5. This "Edge Computing" standard eliminated the risk of patient data interception. As we transition to a Cloud-based architecture in the future, we will adopt TLS/SSL (Transport Layer Security) standards to encrypt data in transit between the hospital and the cloud servers, maintaining strict GDPR compliance.

7 “Post-project” tasks

Several critical steps are required to move toward a professional clinical tool:

- **Database enrichment & Real-world testing:** Following the Ethical Committee's recent authorization to transport labeled bacteria, the database must be enriched with real photos and videos taken via digital microscopes. This is essential to improve the AI's generalization, as the model currently struggles with new, unseen clinical images despite high training scores.
- **Microscope independence:** Future tests must be conducted across different brands and models of microscopes to ensure the AI results remain accurate and independent of the optical equipment used.
- **Hardware integration:** The digital camera and the Raspberry Pi should be integrated into a single, unified casing. This will improve portability and protect the system in a hospital environment.
- **Technical performance optimization:** Further research is needed to understand why the AI inference time varies between images. Additionally, a "video capture option" should be developed to allow real-time analysis instead of static image capture.
- **Cost benchmarking:** A formal comparison must be performed between the ALLEGRI solution's cost and the high-end digital microscopes used at Saint-Denis Hospital to highlight the economic value of our low-cost innovation.
- **Architecture & remote handover:** The GitHub repository must be finalized with comprehensive documentation, including all research notes, source code, and system configurations (IP addresses, remote functioning setup). This ensures the hospital or the next team can maintain and operate the system remotely.

8 Project Completion Recommendations

Based on the execution, testing, and outcomes of the ALLEGRI project across the Hardware, AI, and Software domains, we highlight the following key recommendations for future iterations or handover:

1. Best practices to be generalized :

- **Decentralized, specialized work pairs** : Dividing the team into three autonomous pairs (AI development, hardware/transmission, and web/mobile interface) was highly effective. This prevented technical bottlenecks, allowed parallel workflows, and should remain the standard organizational strategy for this project.
- **Separation of concerns (modular architecture)** : Designing the system where the camera, the processing unit, and the user interface are distinct modules allowed us to easily identify where the system was succeeding (image capture) and where it was bottlenecking (local RAM), which is vital for future troubleshooting.

2. Improvements : methodological and architectural aspects to be reviewed

- **Architectural pivot from Edge AI to Cloud Inference** : Initially, deploying the YOLOv11 model locally on the Raspberry Pi (Edge Computing) was chosen to ensure data privacy. However, real-world testing revealed severe hardware limitations. The 4GB RAM of the Raspberry Pi struggles under the computational load, especially when analyzing slides with a high concentration of bacteria, leading to excessive processing times and occasional system crashes. By contrast, testing the model on a standard laptop CPU was significantly faster. **Recommendation:** Future iterations must migrate the AI processing to a dedicated Cloud service or scalable API infrastructure. The Raspberry Pi should be repurposed strictly as a lightweight image acquisition and transmission hub, sending USB camera captures to the cloud for heavy inference.
- **Addressing Android network instability:** During testing, the Android application encountered frequent connection drops when relying on the local network hosted by the Raspberry Pi. **Recommendation:** Transitioning to a cloud-hosted backend will simultaneously solve this issue, allowing the mobile application to utilize standard, stable internet protocols (4G/5G or established Hospital Wi-Fi) rather than relying on the Pi's limited local network capabilities.
- **Proactive dataset auditing (AI):** The technical time required for data preprocessing vastly exceeded our initial estimates due to extreme discrepancies in public datasets (example : 70% empty backgrounds). Future teams must perform rigorous exploratory data analysis (EDA) before allocating compute resources.

3. The peripheral actions to be taken (corrective actions):

- **Formal data-sharing agreements:** The inability to test the system extensively on real clinical slides due to a lack of formal hospital authorization hindered our ability to optimize the AI. The immediate corrective action is to establish a secure, anonymized data-sharing agreement to acquire real-world validation data.
- **Hardware finalization:** While the current digital camera and Raspberry Pi assembly functions as a proof-of-concept, a corrective action before real clinical deployment is to design and manufacture a robust, medical-grade casing to house the Pi and USB cables securely.

9 Annexes to the closure report.

- Bresser Mikrocarn SF 5.0 MP : https://www.amazon.fr/Bresser-MikroCam-Microscope-Adaptateurs-Logiciel/dp/B0F23KJZ1Y/ref=sr_1_2?mk_fr_FR=%C3%85M%C3%85%C5%BD%C3%95%C3%91&s=electronics&sr=1-2
- AmScope 0.5x reduction lens : https://www.amazon.fr/AmScope-RU050-Objectif-r%C3%A9duction-Microscope/dp/B079MLW5B3/ref=sr_1_1?mk_fr_FR=%C3%85M%C3%85%C5%BD%C3%95%C3%91&s=electronics&sr=1-1
- Github link : [BouchentoufOthman/Allegri-AI](#)
- Some extracts of our organization on Microsoft Teams space :

Gestion de projet | Grille | Tableau | Planning | Graphiques | Filtres | Réorganiser par Compétence

Exécutives	Développement de l'IA	Interface Utilisateur	Récupération d'image	Responsables métier
Exécutives Ajouter une tâche Video Zéro 15/04 100% 100% Choix présentation 15/04 100% 100% Final Report 15/04 100% 100% Tâches terminées 17	Développement de l'IA Ajouter une tâche Génération documents 15/04 100% 100% Terminé par SOUFI Shubham... Intégration images Terminé par CATEAU Camille L. 15/04 100% 100% Triangulation images (data cleaning + preprocessing) Terminé par CATEAU Camille L. 15/04 100% 100% Assignation des notes de travail Terminé par CATEAU Camille L.	Interface Utilisateur Ajouter une tâche Application finale 15/04 100% 100% Tâches terminées 4 Réalisation de la Maquette Terminé par CATEAU Camille L. 15/04 100% 100% Développement frontend Terminé par CATEAU Camille L. 15/04 100% 100% Intégration de... Terminé par CATEAU Camille L.	Récupération d'image Ajouter une tâche Système de blocage 15/04 100% 100% Transmission automatique des images Tâches terminées 7	Responsables métier Ajouter une tâche Responsable... Admission Responsable... Contact

8	☒ Présentation-partenaires entreprises ?	 MERCIER Thoni...  CATTEAU Camille...  SHANN Coralie...  BOLLU Shubha...  BOUCHENTOU...  O'KELLY Maxen...	jeu. 26 mars	Done!
9	☒ Ajouter les coach-soft-skills dans le canal	 O'KELLY Maxen...	dim. 29 mars	Done!
10	☒ Export Soft-skills à déposer !!	 CATTEAU Camille...	dim. 29 mars	Done!
11	☒ Soft skills evaluation - 8h15	Ajouter une personne responsable	jeu. 2 avr.	Done!
12	☐ Report (suivre le template donné)	Ajouter une personne responsable	ven. 10 avr.	À faire
13	☒ Présentation pour la closure defense	Ajouter une personne responsable	ven. 10 avr.	Done!
14	☒ Visuel (logo image ou autre)	 MERCIER Thoni...	ven. 10 avr.	Done!
15	☐ Vidéo - 2min max	 O'KELLY Maxen...	ven. 10 avr.	À faire
16	☐ Abstract français + anglais	Ajouter une personne responsable	ven. 10 avr.	À faire